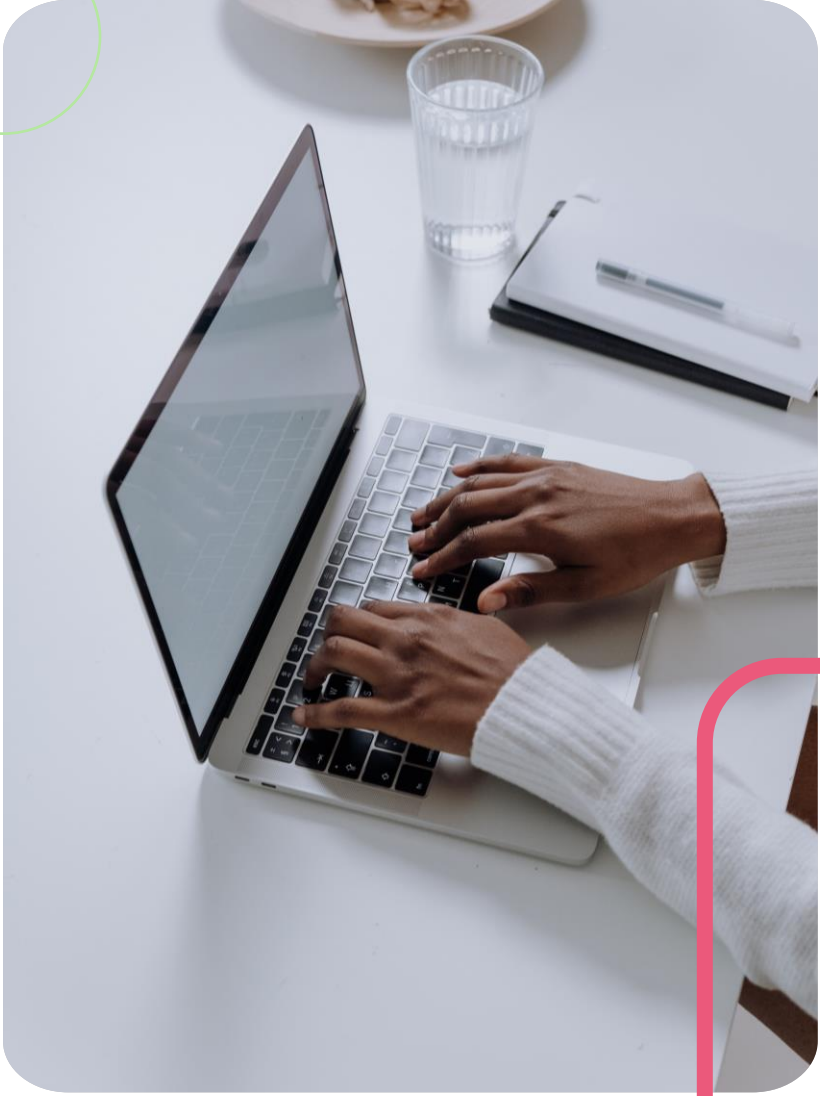




# Diploma in **Data Analytics**

Lesson 7: Multiple linear regression continued



Multiple linear regression continued <

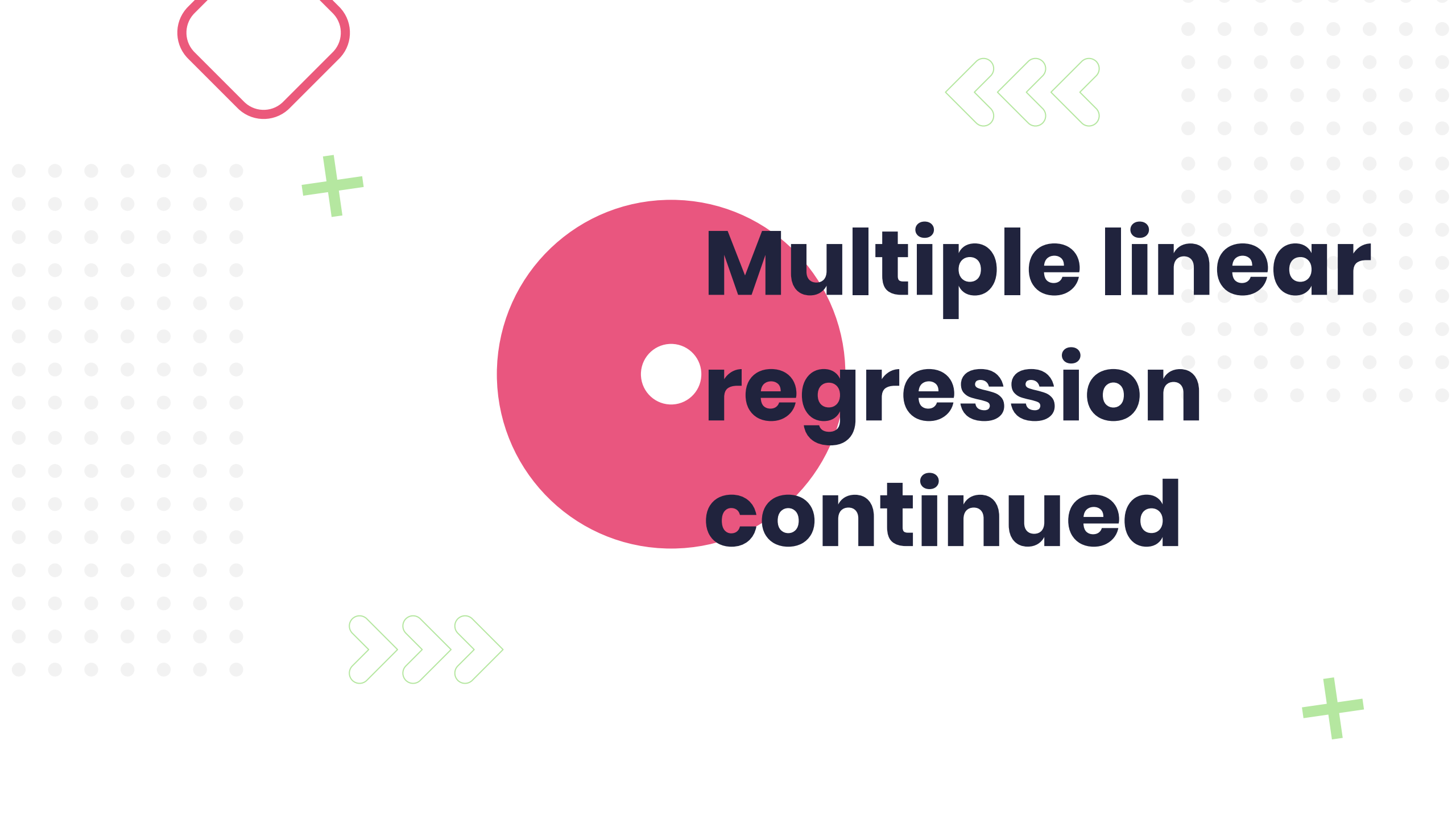
Problems with linear models <

Intro to logistic regression <



# Objectives

Lesson 7



# Multiple linear regression continued

# Recap multiple linear regression

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \varepsilon$$

$y_i$  = dependent variable

$x_i$  = explanatory variables

$\beta_0$  = y-intercept

$\beta_p$  = slope coefficients for explanatory variables

$\varepsilon$  = error term



# Uses of multiple linear regression

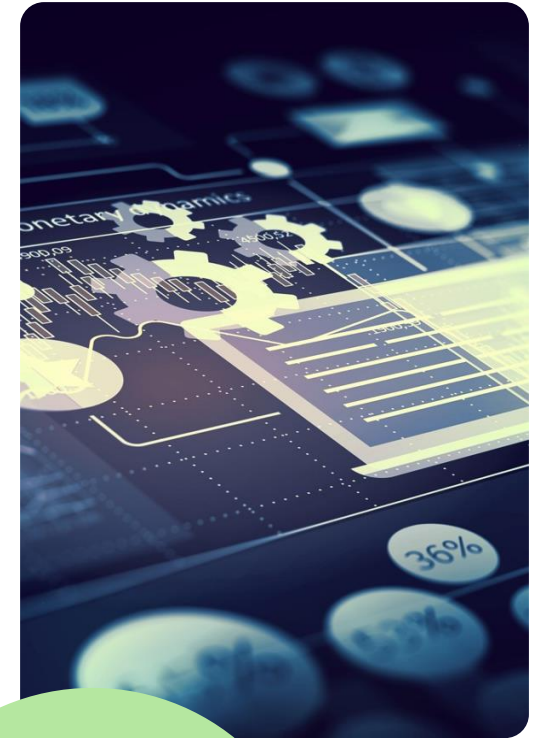
**Forecast effects**



**Identify strength  
of effect**



**Predict trends**





# Assumptions of multiple linear regression

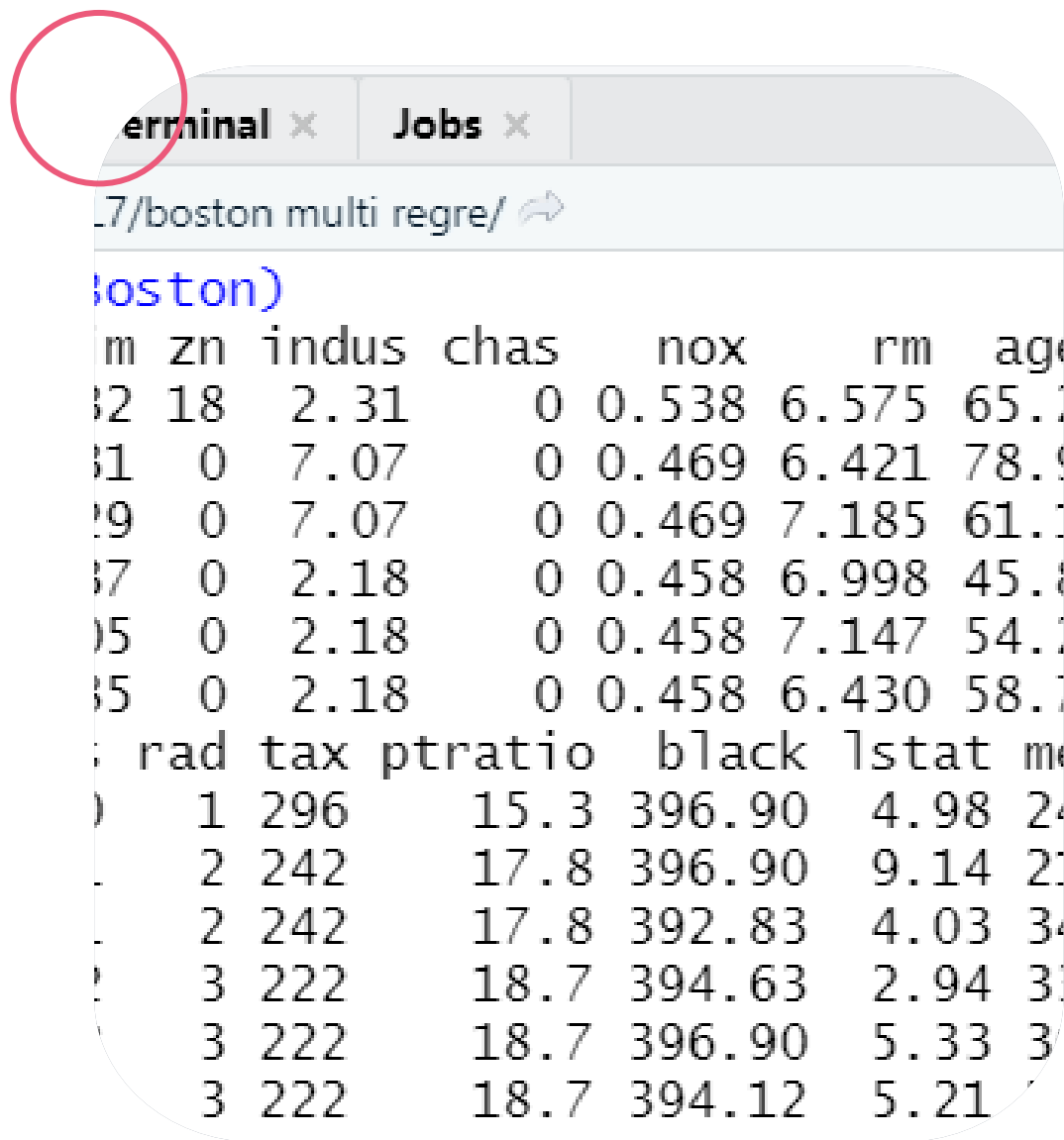
- Linear relationship between dependent & independent variables
- Independent variables are not highly correlated
- Independent variables are independent and randomly sampled
- Residuals mean = 0, variance =  $\sigma$



# Importance of variables

- F-statistic
- P-value
- Variable selection
  - Forward selection
  - Backward selection
  - Mixed selection



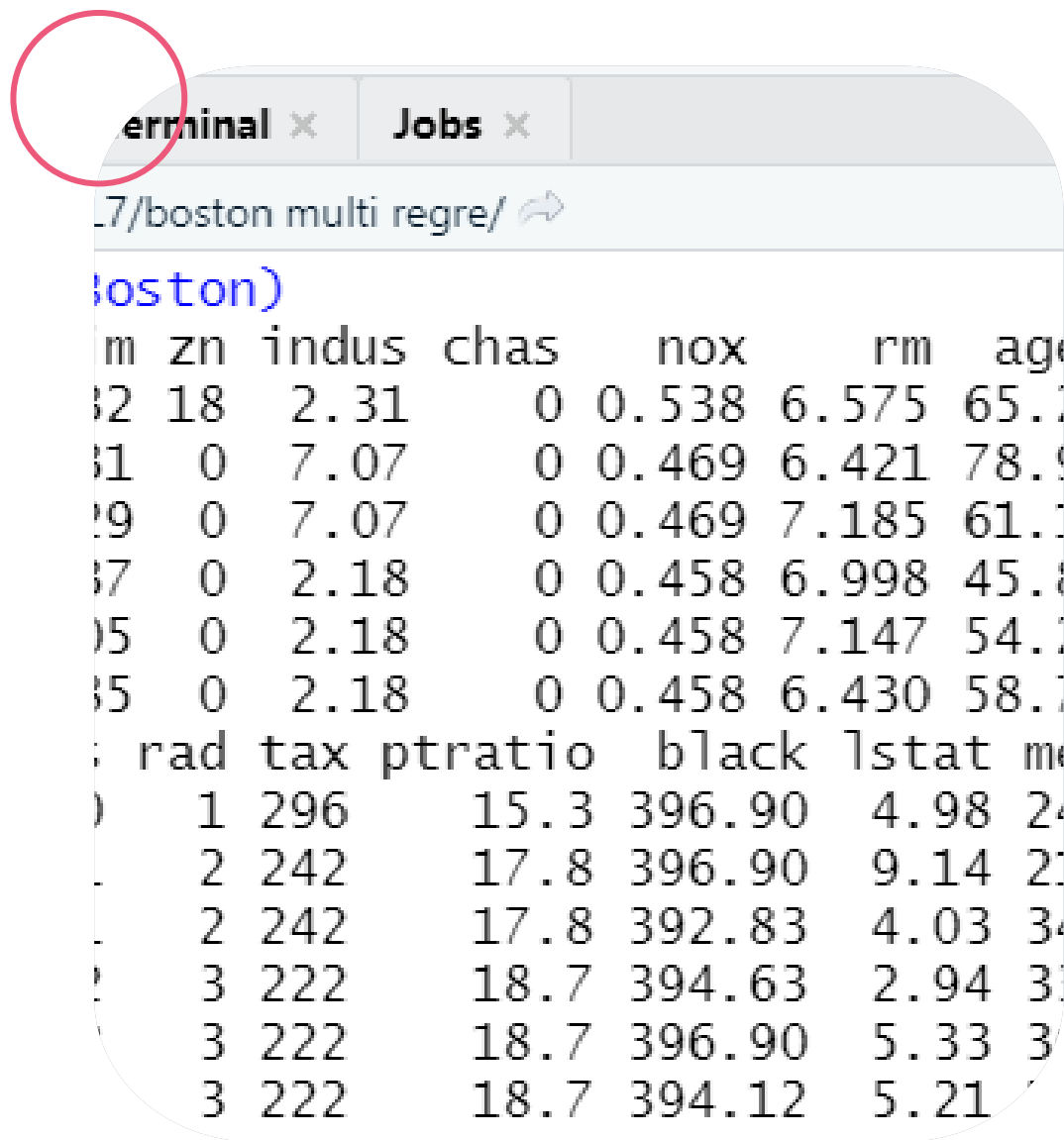


|    | rm  | zn   | indus   | chas   | nox   | rm   | age |
|----|-----|------|---------|--------|-------|------|-----|
| 32 | 18  | 2.31 | 0       | 0.538  | 6.575 | 65.1 |     |
| 31 | 0   | 7.07 | 0       | 0.469  | 6.421 | 78.9 |     |
| 29 | 0   | 7.07 | 0       | 0.469  | 7.185 | 61.1 |     |
| 37 | 0   | 2.18 | 0       | 0.458  | 6.998 | 45.8 |     |
| 05 | 0   | 2.18 | 0       | 0.458  | 7.147 | 54.1 |     |
| 35 | 0   | 2.18 | 0       | 0.458  | 6.430 | 58.1 |     |
|    | rad | tax  | ptratio | black  | lstat | me   |     |
| 0  | 1   | 296  | 15.3    | 396.90 | 4.98  | 24   |     |
|    | 2   | 242  | 17.8    | 396.90 | 9.14  | 21   |     |
|    | 2   | 242  | 17.8    | 392.83 | 4.03  | 34   |     |
|    | 3   | 222  | 18.7    | 394.63 | 2.94  | 31   |     |
|    | 3   | 222  | 18.7    | 396.90 | 5.33  | 31   |     |
|    | 3   | 222  | 18.7    | 394.12 | 5.21  | 31   |     |

## Forward selection

- Starts with a small model
- Adds variables at each step of the process until criterion stops improving

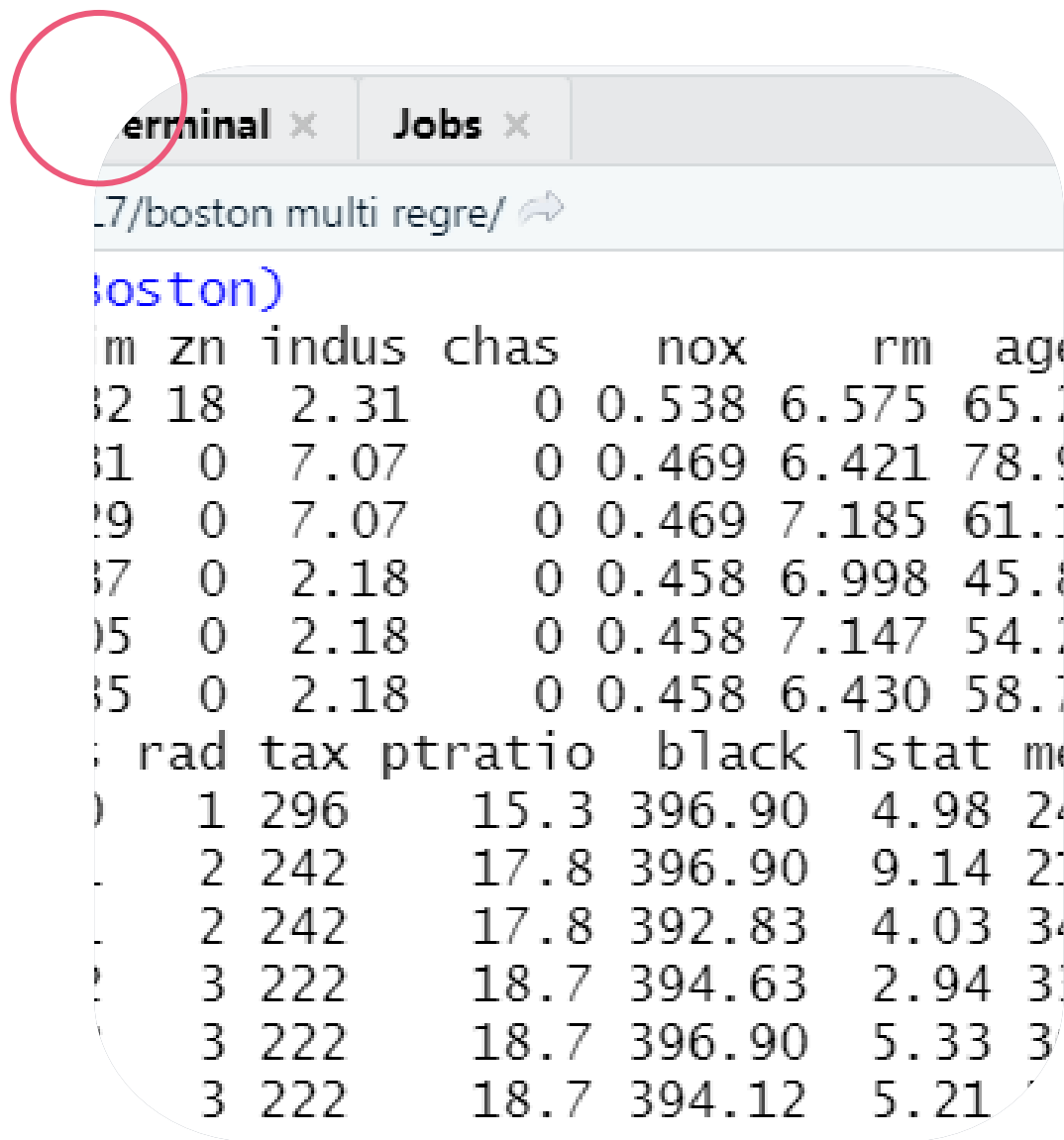




```
terminal x Jobs x
_7/boston multi regre/ ↵
boston)
rm zn indus chas nox rm age
32 18 2.31 0 0.538 6.575 65.
31 0 7.07 0 0.469 6.421 78.
29 0 7.07 0 0.469 7.185 61.
37 0 2.18 0 0.458 6.998 45.
05 0 2.18 0 0.458 7.147 54.
35 0 2.18 0 0.458 6.430 58.
rad tax ptratio black lstat me
0 1 296 15.3 396.90 4.98 24
2 242 17.8 396.90 9.14 25
2 242 17.8 392.83 4.03 34
3 222 18.7 394.63 2.94 38
3 222 18.7 396.90 5.33 38
3 222 18.7 394.12 5.21 38
```

## Backward selection

- Starts with model that contains all the variables
- Removes a variable at each step until predetermined criterion is satisfied



|    | rm  | zn   | indus   | chas   | nox   | rm   | age |
|----|-----|------|---------|--------|-------|------|-----|
| 32 | 18  | 2.31 | 0       | 0.538  | 6.575 | 65.1 |     |
| 31 | 0   | 7.07 | 0       | 0.469  | 6.421 | 78.9 |     |
| 29 | 0   | 7.07 | 0       | 0.469  | 7.185 | 61.1 |     |
| 37 | 0   | 2.18 | 0       | 0.458  | 6.998 | 45.8 |     |
| 05 | 0   | 2.18 | 0       | 0.458  | 7.147 | 54.1 |     |
| 35 | 0   | 2.18 | 0       | 0.458  | 6.430 | 58.1 |     |
|    | rad | tax  | ptratio | black  | lstat | medv |     |
| 0  | 1   | 296  | 15.3    | 396.90 | 4.98  | 24.1 |     |
| 1  | 2   | 242  | 17.8    | 396.90 | 9.14  | 21.1 |     |
| 2  | 2   | 242  | 17.8    | 392.83 | 4.03  | 34.1 |     |
| 3  | 3   | 222  | 18.7    | 394.63 | 2.94  | 31.1 |     |
| 4  | 3   | 222  | 18.7    | 396.90 | 5.33  | 31.1 |     |
| 5  | 3   | 222  | 18.7    | 394.12 | 5.21  | 31.1 |     |

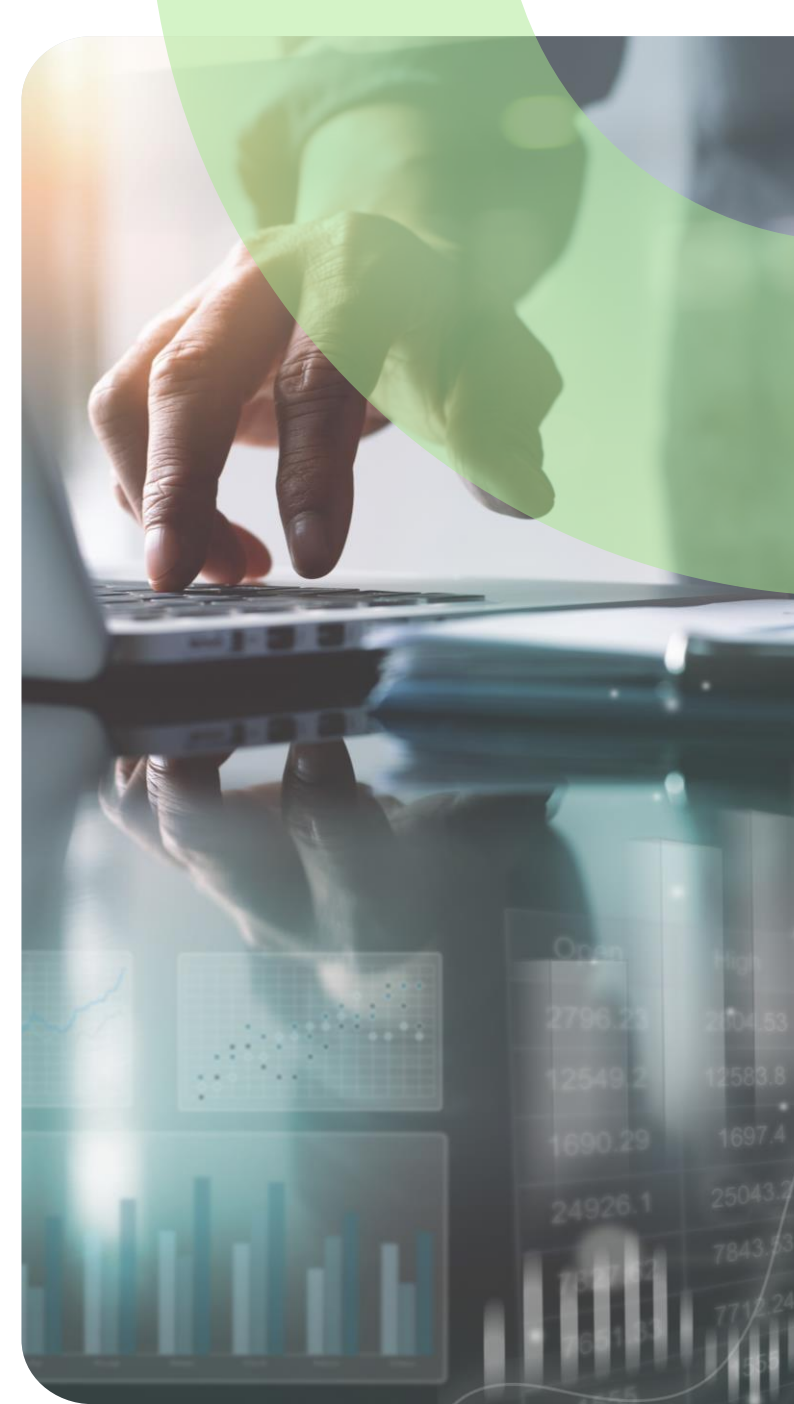
## Mixed selection

- Combination of forwards and backwards selection process
- Addresses a foundational drawback of the forward selection
- Smaller model than backward selection



# Model fit statistics

- $R^2$ 
  - Coefficient of determination
  - Measures variation in the outcome explained by independent variables
- Residual standard error
  - Average amount the response deviates from the actual regression line

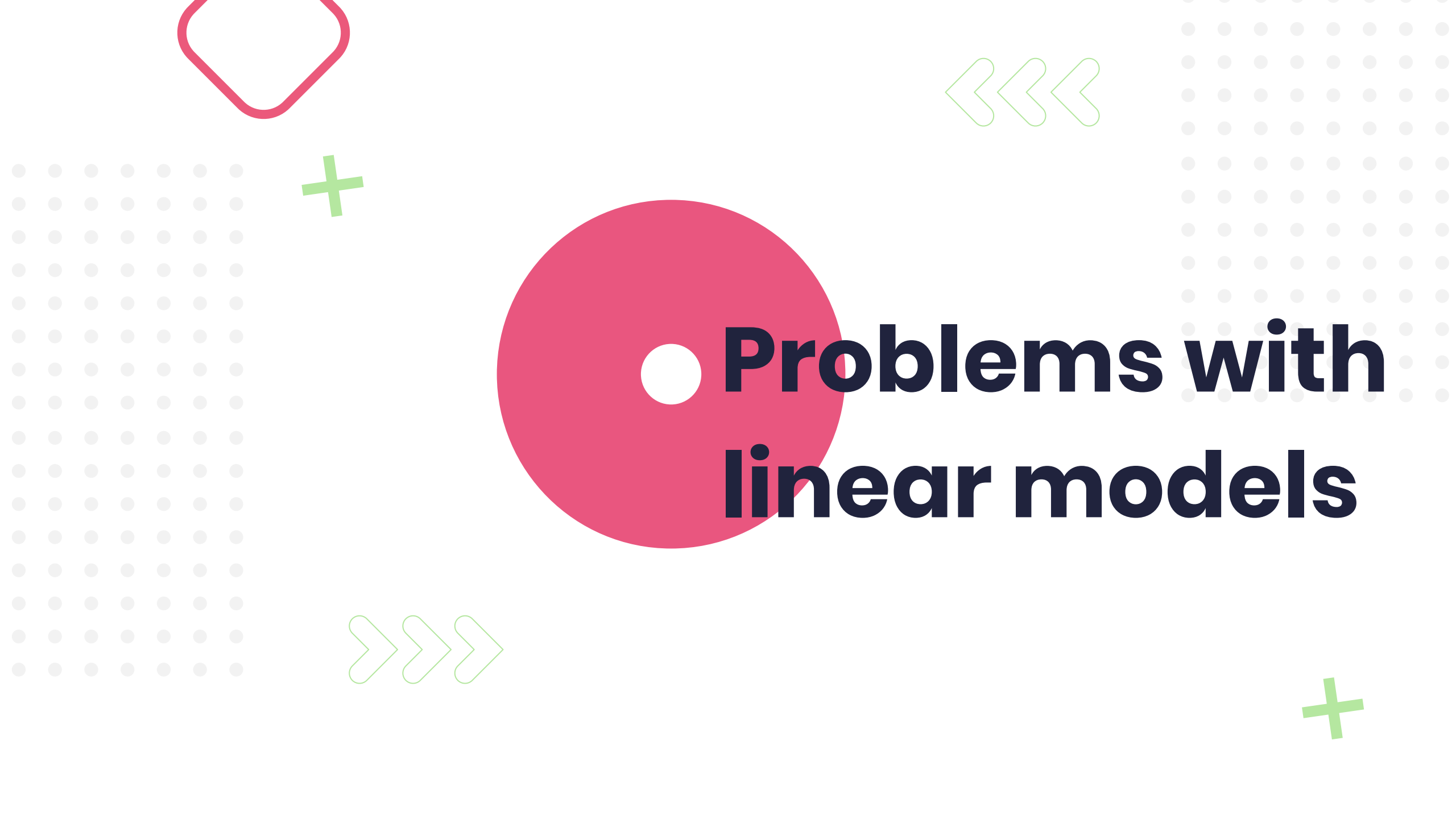


# More model fit statistics

- Mallow's  $C_p$
- Akaike information criterion (AIC)
- Bayesian information criterion (BIC)
- Adjusted Akaike information criterion
- Adjusted Bayesian information criterion
- Adjusted  $R^2$







# Problems with linear models



# Problems with linear regression model

- Non-linearity
- Correlation between error terms
- Non-constant variance of error terms
- Outliers
- High leverage points
- Collinearity

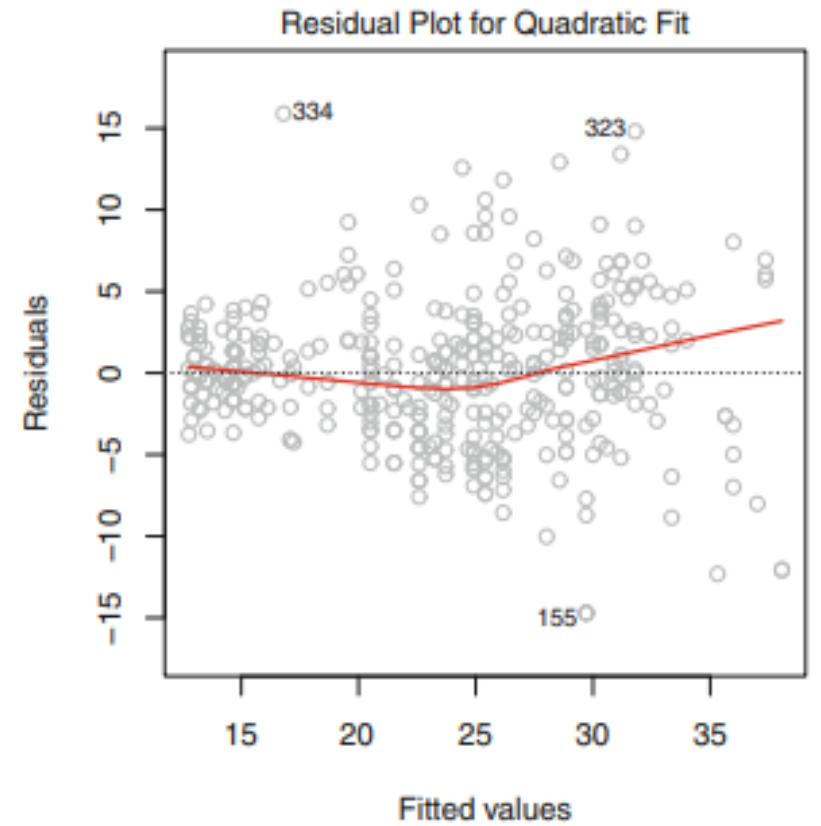
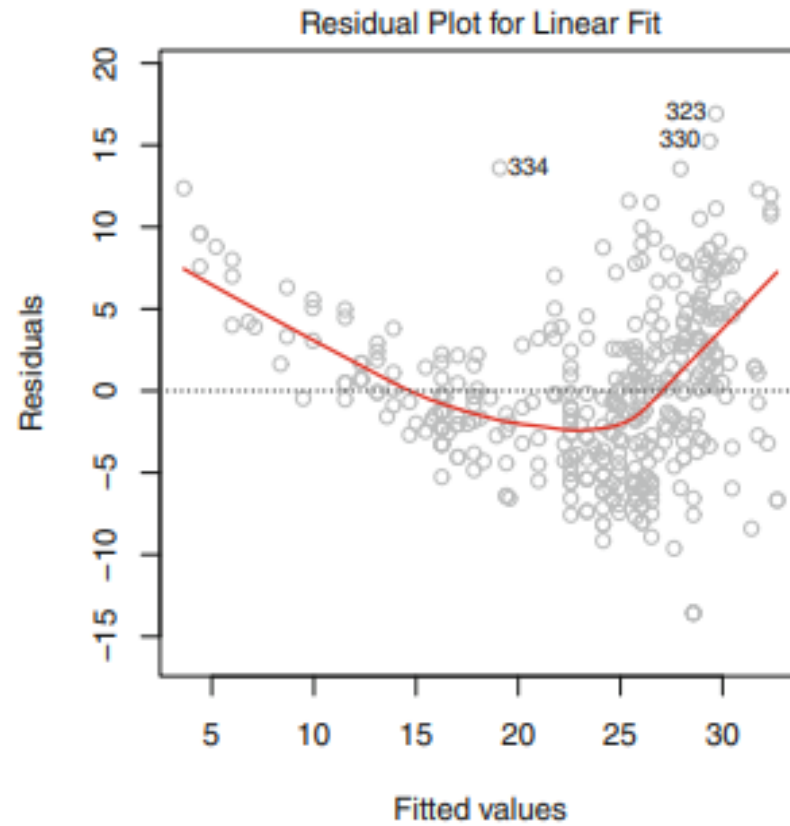
# Non-linearity

If relationship between independent and dependent variable is actually non-linear, accuracy of model is decreased



# Residual plots

Linear regression residual plot shows a pattern



# Correlation of error terms

- Error terms are uncorrelated
- Example:  $\varepsilon_i$  positive inclination provides no information about  $\varepsilon_j$



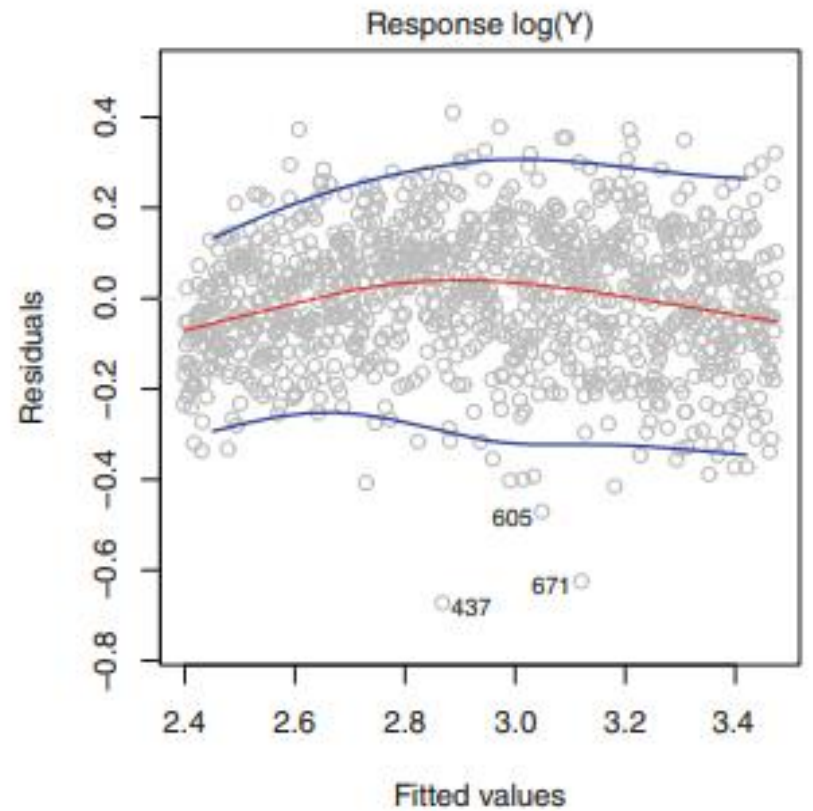
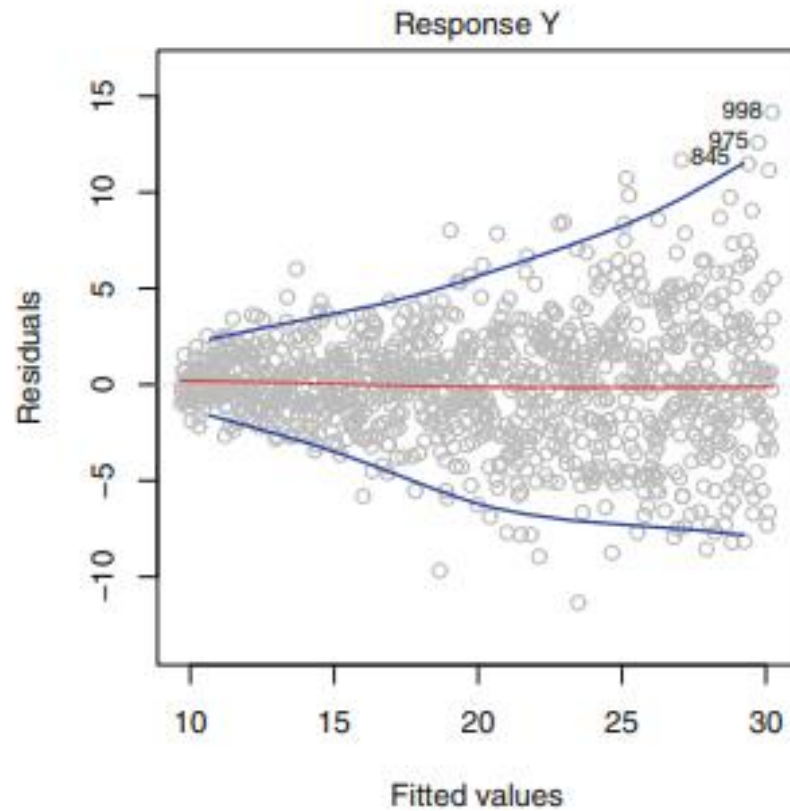


# Non-constant variance of error terms

- Linear regression model should have constant variance of error terms,  $\sigma^2$
- Visual way to check constant variance, is a residual versus fitted plot



# Residuals vs fitted plot



Source: James et al., 2013

# Outliers

- Data point that varies significantly from other data points
- Causes can include incorrect recordings



# High leverage points

- Extreme high or low value for  $x_i$
- Observation considered an outlier *iff* values are extreme w.r.t. other y-values





# Collinearity

- Two or more independent variables that are closely related
- Decreases reliability of estimates of regression coefficients







# Intro to logistic regression

# Why not linear regression

- Qualitative response variable is not applicable for linear regression
- $Y = \begin{cases} 1 & \text{if Facebook;} \\ 2 & \text{if Twitter;} \\ 3 & \text{if Instagram.} \end{cases}$

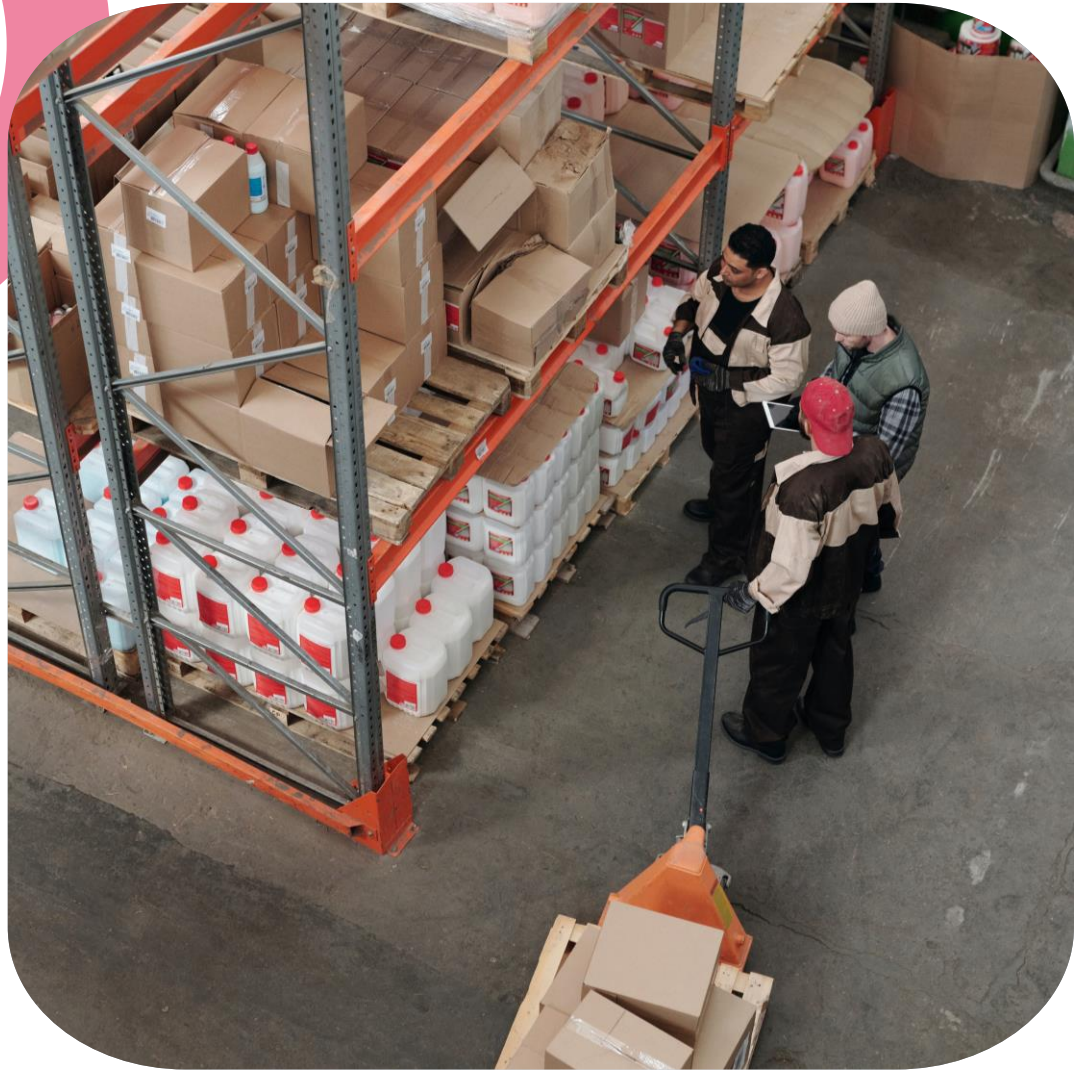


# Logistic regression

- True or false outcome
- Logistic regression fits an s-shape function to data







# Logistic regression in R

- `glm()`
- `family = binomial`



# Challenge >>

Determine if the linear model is a good fit for the Boston data set

**#exploredata**